

Course Outline

Course/Class Information

Faculty/School Name	Mathematical and Computational Sciences	Academic Year and Semester	Fall 2026
Course Name	Theory of Computing	Course Code	MCS – 3320
Credits	3	Pre-requisites	CS 2910, CS 2920 and MATH 2420 - Must be completed prior to taking this course.
Class Time	Monday 13:35 - 14:50 Thursday 09:50 - 11:05	Location	A510
Passing Grade/ Prerequisite Grade	50% / 50%		

Instructor and TA Contact Information

Instructor Name	Hossam Mekawy	E-mail	Hossam.Mekawy@uofcanada.edu.eg
Office	B211	Office Hours	Monday 9:00 – 11:00
TA Name	Omar Eltouny	E-mail	Omar.ayman@uofcanada.edu.eg
Office	B318	Office Hours	Sunday, Monday: 9-10 am

Course Description

This course introduces automata theory, formal languages and computability. Topics include: finite automata; regular expressions; regular, context-free, and context-sensitive languages; computability models; algorithmic decidable and undecidable problems. Three lecture hours per week

Course Learning Objectives

Upon successful completion of this course, students will be able to:

1. Understand why studying Automata theory, complexity theory and computability theory and their possible applications
2. Understand proof methods and how they are utilized for proving several properties of languages, grammar, and automata.
3. Ability to design (non-) deterministic Finite Automata (DFA, NFA), regular expressions, Pushdown Automata (PDA), Context-free Grammars (CFGs), and Turing Machines (TMs).
4. Convert among DFAs, NFAs, and regular expressions. Also, between PDAs and CFGs.
5. Describe possible applications for NFA, regular expressions, context free languages, and TMs.
6. Explain the meaning of the halting problem for Turing machines and its undecidability.
7. Explain the classification of problems based on their computational time complexity (i.e., P, NP, and NP-complete).
8. Demonstrate understanding of the above by coding applicable programs and solving exercises.

Assessment of Learning Objectives

The learning objectives of this course are assessed and achieved as follows:

Assessment Method Of Learning Objective	Examinations	Assignment	Class Activities and Participation
L.O. 1	Quiz1, Midterm, Final	Assignment 1.	yes
L.O. 2	Quiz1, Midterm, Final	Assignment 1.	yes
L.O. 3	Quiz 1, Midterm, Final	Assignment 1	yes
L.O. 4	Quiz 2, Midterm, Final	Assignment 1, Assignment 2	yes
L.O. 5	Quiz 2, Midterm, Final	Assignment 1, Assignment 2, Assignment 3,	yes
L.O. 6	Quiz 2, Final,	Assignment 3	yes
L.O. 7	Final	Assignment 3,	yes
L.O. 8	Midterm, Quiz1, Quiz 2, Final	Assignment 1, Assignment 2,	yes

Assignment 3,

Required Textbook & Course Material

Textbook

- Introduction to Automata Theory, Languages, and Computation 3rd Edition, by John E. Hopcroft, Rajeev Motwani, and Jeffrey D. Ullman.
- Additional reading:
Sipster, Michael. Introduction to the Theory of Computation. 2nd ed. Boston, MA: Thomson Course Technology, 2006. ISBN: 0534950973.

Lecture Slides (PPTs.) on the D2L LMS

Grading Scheme

<i>Assessment</i>	<i>Assessment</i>	<i>Weight</i>	<i>Date</i>
1	Lecture Participation	15%	All lectures
2	Quiz 1	5%	Week3: Oct 15 th
3	Assignment 1	10%	Week4: Oct 22 nd
4	Midterm	15%	Week5: Oct 29 th
5	Assignment 2	10%	Week7: Nov 12 th
6	Quiz 2	5%	Week9: Nov 26 th
7	Assignment 3	10%	Week11: Dec 10 th
	Final Exam	30%	All lectures
Total		100	

Grading Scale

UPEI grade scale is as follows:

Grade	GP	Percentage Range	Verbal Descriptor
A+	4.3	91% - 100%	Exceptional
A	4.0	85% - 90%	Excellent
A-	3.7	80% - 84%	Extremely good

B+	3.3	77% - 79%	<i>Very good</i>
B	3.0	74% - 76%	<i>Good</i>
B-	2.7	70% - 73%	<i>Fairly good</i>
C+	2.3	67% - 69%	<i>More than adequate</i>
C	2.0	64% - 66%	<i>Adequate</i>
C-	1.7	60% - 63%	<i>Minimum for good Academic standing</i>
D+	1.3	57% - 59%	<i>Below good standing</i>
D	1.0	54% - 56%	<i>Minimally acceptable</i>
D-	0.7	50% - 53%	<i>Barely acceptable</i>
F	0.0	0% - 49%	<i>Below university requirements</i>

Class Schedule

<i>Week</i>	<i>Lecture</i>	<i>Reading Material (if applicable)</i>
1	Introduction to Theory of Computation: Motivation - Finite Automata introduction - Proof Methods- Quine exercise	Chapter 1
2	DFA -NFA -NFA and DFA equivalence - NFA to DFA conversion - FA with epsilon transition	Chapters 2
3	Regular expressions and languages: operators - Building regular expressions - precedence - finite automata to regular expressions and vice versa- applications	Chapter 3
4	Regular languages properties: closure properties - decision properties - equivalence and minimization of DFA.	Chapter 4
5	Midterm	
6	Context-free grammar (CFG)-Parse Trees- Applications	Chapter 5
7	Pushdown Automata (PDA): Definition - acceptance types - Equivalence with CFG - deterministic PDA	Chapter 6
8	CFL properties: Simplify - closure and decision properties	Chapter 7
9	Turing Machine, its programming techniques, simulating a computer using a TM and vice versa.	Chapter 8
10	Undecidability - recursive - recursively enumerable	Chapter 9
11	the halting problem	Chapter 9
12	Intractable and unsolvable problems	Chapter 10



University Policies

Attendance and Punctuality

Students are required to attend all classes and be on time for all classes. There are no excuses for missing class. All absences are counted as time missed. This applies to all lectures, tutorials / labs and mandatory office hours for probation & Gateway students. Accordingly, students should note the following:

1. The instructors are not required to substitute assignments or to make up examinations for absent students.
2. It is the student's responsibility to make up the class content missed in consultation with his/her instructor(s). This does not mean that the instructor will provide a substitute class for the student.
3. Grades can be affected by missing a class as the instructor may include class participation marks and/or in-class assignments, or pop-up quizzes as part of the assessment plan for the course.
4. If a student is absent for 25% or more of the class time, tutorials, labs, and/or mandatory office hours, the instructor may:
 - a. Advise the student to drop the course if the absences occur before the drop date for a course.
 - b. Give the student a grade of F for the course
 - c. Give the student an Incomplete grade for the course if the absences are caused by an approved documented illness or emergency.

Code of Conduct for Students

The University of Prince Edward Island expects all students to comply with all University regulations and policies, all government laws, and any codes of ethics that govern students who are members of regulated professions. All students are strongly advised to become familiar with this code, which can be accessed at: [UofCanada Student Code of Conduct \(Non-Academic\).pdf](#)

Academic Integrity

All students are expected to conduct themselves in all aspects of this course with the highest level of academic integrity. Accordingly, academic dishonesty will not be tolerated. Regulation 20 of the University Calendar defines academic dishonesty. Where there is reasonable evidence to support an allegation of academic dishonesty, Regulation 20 will be strictly enforced. All students are strongly advised to become familiar with this Regulation and UPEI Cairo Academic Integrity policy which can be accessed at: [UPEI Cairo Academic Integrity Policy.pdf](#)

Intellectual Property

The teaching materials used in this course are the intellectual property of the instructor. To avoid copyright violations, students must not share, distribute, or transmit any course materials including text files, video files, or other content to any third party without prior written permission.

The Use of AI

The use of generative artificial intelligence (AI) and AI-assisted technologies is permitted only upon receiving written approval from the instructor. Students should disclose in their

assignments the use of AI and AI-assisted technologies in the writing process. Failure to do so will be considered academic dishonesty.

Student Accommodation

The university strives to make the learning experience as accessible and inclusive as possible. If you have accessibility needs that require academic accommodation, kindly contact the student wellness department and review UPEI Cairo campus academic accommodation policy, which can be accessed at: [UPEI Cairo Campus Academic Accommodations Policy.pdf](#)

TEACHING AND COMMUNICATIONS

The time that we spend together in the classroom is vital to be able to fully understand the course material in the textbook complemented with discussion and activity in the classroom. Activities may include in-class presentations, case studies, issue discussions, simulations, and problem solving. These activities may be done either individually, as a group, or as a class. The expectation is that all students will engage their instructor and fellow students in these activities, both respectfully and professionally.

Students must check the learning management system and university email regularly for confirmation on assignments and announcements. **It is YOUR responsibility to be knowledgeable of the latest developments in the course.**

Expectations From Students

1. There will be no make-up exams allowed without a valid reason or clinic approval.
2. All reports required for this course must be original and adhere to the university's academic integrity policy. Reports will be submitted through Turnitin, which checks for similarities with other submissions and online sources. If similarity beyond a reasonable amount is detected, penalties extending to filing an academic dishonesty case may be applied.
3. All students are expected to submit their work on time; late submissions without a prior approved excuse will not be accepted.
4. Students who arrive late at class after 10 minutes will not be allowed entrance and will be marked absent.
5. In keeping with classroom best practices for creating an engaging learning environment, all mobile phones must be turned off or on “vibrate” mode.
6. Once you come to class, it is expected that you will be there to stay. Please make sure you use the washroom before or after class.
7. All students agree to abide by the following professional standards:
 1. Come to class on time and be prepared
 2. If unable to attend class, contact the instructor/students before the class
 3. Be considerate and respectful of others’ time, opinions, and rights
 4. Attend to what is happening in class – do not text/surf/use cell phone or use

- electronic devices for non-class purposes
5. Seek and provide feedback to/ from peers and instructors.
 6. Commit to the highest standards of competence and academic integrity (work completed on time, motivated, dress appropriately etc.)
 7. Contribute to teamwork in a fair and balanced manner.

WHILE EVERY EFFORT WILL BE MADE TO FOLLOW THE SCHEDULE, THE GRADING SCHEME AND THE ASSESSMENT DATES. PLEASE NOTE THAT UNFORESEEN CIRCUMSTANCES CAN LEAD TO SOME CHANGES.